Numerical Methods
LAB 2

1. Consider a simple floating point representation:

$$rd(x) = (-1)^k(0.d_1d_2)2^(e-b)$$

where $(-1)^k$ (k - a single bit) defines a sign, 2 bits are used for mantissa (d_1, d_2), e is represented by 2 bits and $b = 1$. Write down all possible real numbers represented in this system. What is the largest and the smallest number?

2. Write a code in python (or other language) calculating $fl(\epsilon + 1)$. By setting epsilon $= 1/2^n$ and increasing n find, when $\epsilon + 1$ starts to get 1. Use various precisions: *numpy.float16* (16 bits), *numpy.float32* (32 bits), *numpy.float64* (64 bits) and *numpy.float128* (128 bits).

   Hint:

   The code below loops over $n = (0, ..., 10)$ and prints n in 16-bit precision:

```
import numpy as np

x=0
n=0
while x<10:
    x=(1+np.float16(n))
    print("n = ",n," x = ",x)
    n+=1
```

3. The $\sin(x)$ and cos can be approximated by a Tylor series:

$$sin(x) = \sum_{n=0}^{\infty}(-1)^n \cdot \frac{x^{2n+1}}{(2n+1)!}$$

$$cos(x) = \sum_{n=0}^{\infty} (-1)^n \cdot \frac{x^{2n}}{(2n)!}.$$

Write and test a code in python (or in any other language) that calculates or better plots the the following:

- sin from *numpy* package compared to the sin approximated by n terms of a Tylor series,
- the same for cosine,
- plot or print a difference between approximated sin or cos and the corresponding function from the *numpy* package
- print or plot $\sin^2(x) + \cos^2(x) - 1$ for trygonometric functions approximated by n terms of a Tylor series.

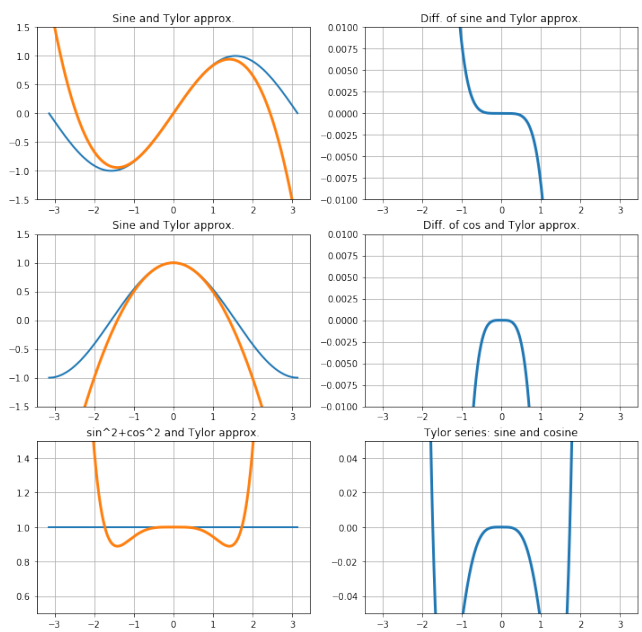

Figure 1: Output plot from your code

-

Hints:

- Code to calculate and draw function sin in python. It can be used as a starting point for your code:

```
import numpy as np
import matplotlib.pyplot as plot
```

```
# Get x values of the sine wave
time          = np.arange(0, 10, 0.1);

# Amplitude of the sine wave is sine
# of a variable like time
amplitude     = np.sin(time)

# Plot a sine wave using time and amplitude
# obtained for the sine wave
plot.plot(time, amplitude)


# Give a title for the sine wave plot
plot.title('Sine wave')

# Give x axis label for the sine wave plot
plot.xlabel('Time')
# Give y axis label for the sine wave plot
plot.ylabel('Amplitude = sin(time)')


plot.grid(True, which='both')
plot.axhline(y=0, color='k')

plot.show()



# Display the sine wave

plot.show()
```

- Use Google Colab and create your own github archive *https://github.com/* to share the code with others.
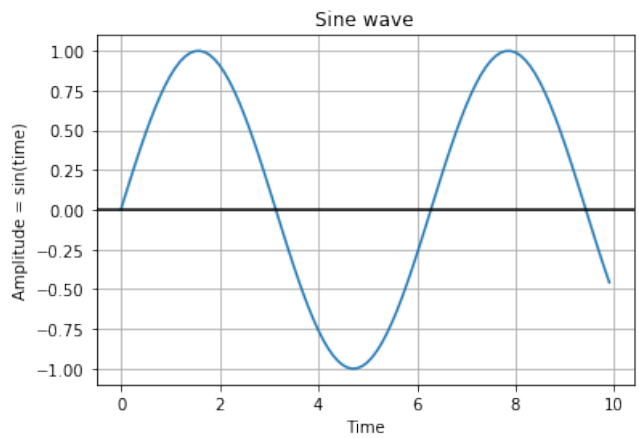
Figure 2: Output from the above code: